Recovering AEWR Sampling-Composition Shocks from the Farm Labor Survey

1. Motivation

The Adverse Effect Wage Rate (AEWR) is based on the USDA National Agricultural Statistics Service (NASS) Farm Labor Survey (FLS). The FLS is a probability survey rather than a census. Consequently, the wage estimate used to set the AEWR depends partly on which farms are selected, which farms respond, and the survey weights assigned to the usable responses in a given year.

This creates a potential source of quasi-random variation in the AEWR. Even holding fixed the underlying distribution of agricultural wages, different realizations of the FLS sample can place different effective weight on high- and low-wage portions of an AEWR region. The resulting sampling-composition error changes the regional FLS wage and therefore the subsequent AEWR.

The objective is to recover, or approximate, the component of the AEWR generated by idiosyncratic year-to-year variation in the realized geographic composition of the FLS sample. This is distinct from simply constructing weights that predict the AEWR well. The intended estimand is the random deviation of realized FLS representation from its expected allocation across the constituent parts of an AEWR region.

2. FLS Sampling and Estimation

The FLS uses a dual-frame design consisting of a list-frame sample and an area-frame nonoverlap sample. The list-frame sample is selected using a hierarchical stratified design. Strata are defined primarily by farms' peak number of hired workers or calculated value of sales, with larger operations selected at higher rates. The area-frame component represents farms missing from the list frame.

Each sampled operation initially receives a weight related to the inverse of its probability of selection. Following data collection, weights are adjusted for unit and item nonresponse within state-level sampling strata. Published wage rates are ratio estimates constructed from expanded wage and labor-hour totals. Thus, a farm's effective contribution to the regional wage depends on its selection probability, response status, final survey weight, reported employment and hours, and wage.

Counties are not themselves the primary FLS sampling strata. Nevertheless, random geographic representation arises because the farms selected from each state-size stratum are located in particular counties, and because response patterns can differ across the selected farms. The realized effective weight placed on a county is therefore an aggregation of farm-level sampling and response outcomes.

Let the effective contribution of county i to the FLS wage in AEWR region r and year t be approximately

$$\omega_{irt} \propto \sum_{f \in i} I_{ft} R_{ft} d_{s(f)t} H_{ft},$$

where I_{ft} indicates that farm f was selected, R_{ft} indicates that it supplied a usable response, $d_{s(f)t}$ is its final expansion and nonresponse weight within sampling stratum $s(f)$, and H_{ft} is its reported worker hours. After normalization, the county contributions sum to one within an AEWR region-year.

3. The Sampling-Composition Shock

Let w_{irt} denote the population agricultural wage associated with county i. A simplified representation of the published FLS regional wage is

$$\widehat{W}_{rt}^{FLS} = \sum_{i \in r} \omega_{irt} w_{irt}.$$

Decompose the realized effective county weight into its expected component and a sampling deviation:

$$\omega_{irt} = q_{irt} + u_{irt},$$

where q_{irt} is the county's expected contribution to the regional wage estimator and u_{irt} is the deviation caused by the realized sample and response pattern. Because both sets of weights sum to one,

$$\sum_{i \in r} u_{irt} = 0.$$

The FLS wage can then be written as

$$\widehat{W}_{rt}^{FLS} = \sum_{i \in r} q_{irt} w_{irt} + \underbrace{\sum_{i \in r} u_{irt} w_{irt}}_{\eta_{rt}},$$

where

$$\eta_{rt} = \sum_{i \in r} u_{irt} w_{irt}$$

is the regional sampling-composition shock. A positive value means that the realized sample placed unusually high effective weight on higher-wage portions of the region; a negative value means that it placed unusually high effective weight on lower-wage portions.

This scalar shock, rather than the complete vector of county weights, is the ultimate object of interest for the AEWR design.

4. Expected County Contributions

The starting weights are constructed using BEA farm employment. For county i in region r and year t, define

$$q_{irt} = \frac{E_{it}}{\sum_{j \in r} E_{jt}},$$

where E_{it} is BEA farm employment.

These weights should be interpreted as a proxy for expected effective contribution to the regional wage estimator, not as the literal expected share of questionnaires mailed to the county. The FLS samples farms rather than workers and oversamples operations in large-worker or high-sales strata. Its wage estimator also weights observations through reported worker hours. BEA farm employment is nevertheless a useful starting approximation because counties with more farm employment should generally make larger contributions to a regional agricultural wage.

Alternative starting weights can combine BEA employment with agricultural land from the Cropland Data Layer. This reflects the FLS's use of both a list frame and an area-frame nonoverlap sample. Such a mixture should be treated as a sensitivity specification unless the list- and area-frame shares can be recovered directly for each survey year.

5. Entropy Tilting as an Inverse Problem

County-level OEWS agricultural wages provide an external proxy for w_{irt}. Let x_{it} denote the OEWS wage measure constructed to resemble the FLS field-and-livestock wage. The proposed calibration finds realized county weights that reproduce the published FLS wage while departing as little as possible from the BEA prior.

For each AEWR region-year, solve

$$\min_{\{\omega_{irt}\}} \sum_{i \in r} \omega_{irt} \log \left(\frac{\omega_{irt}}{q_{irt}} \right)$$

subject to

$$\sum_{i \in r} \omega_{irt} = 1$$

and

$$\sum_{i \in r} \omega_{irt} x_{it} = \widehat{W}_{rt}^{FLS}.$$

With one wage constraint, the solution is an exponential tilt of the prior:

$$\omega_{irt}(\lambda_{rt}) = \frac{q_{irt} \exp(\lambda_{rt} x_{it})}{\sum_{j \in r} q_{jrt} \exp(\lambda_{rt} x_{jt})},$$

where λ_{rt} is chosen so that the calibrated OEWS wage equals the FLS target.

The tilting parameter has a useful interpretation:

- $\lambda_{rt} = 0$ means that the BEA allocation already reproduces the FLS wage.
- $\lambda_{rt} > 0$ means that the FLS estimate behaves as if higher-wage counties received greater effective representation than implied by the BEA prior.
- $\lambda_{rt} < 0$ means that the FLS estimate behaves as if lower-wage counties received greater effective representation.

The implied county deviations are

$$\hat{u}_{irt} = \hat{\omega}_{irt} - q_{irt}.$$

6. What Entropy Tilting Does and Does Not Recover

One published regional wage supplies only one scalar restriction on a vector containing many county weights. The realized county weights are therefore not nonparametrically identified. Infinitely many weight vectors can reproduce the same FLS wage.

Entropy tilting selects one of these vectors: the vector closest to the BEA prior under a Kullback-Leibler divergence criterion. The resulting weights should therefore be described as **AEWR-implied geographic calibration weights** or **minimum-divergence implied weights**, rather than the actual realized FLS sampling weights.

With exact calibration, the estimated aggregate shock is mechanically

$$\hat{\eta}_{rt} = \sum_{i \in r} (\hat{\omega}_{irt} - q_{irt}) x_{it} = \widehat{W}_{rt}^{FLS} - \sum_{i \in r} q_{irt} x_{it}.$$

Thus, entropy tilting does not create independent information about the aggregate shock. It provides a disciplined method for allocating the observed FLS-OEWS discrepancy across counties.

Exact calibration is feasible only when the FLS target lies within the convex hull of the observed county OEWS wages. With a single wage constraint, this requires

$$\min_{i \in r} x_{it} \leq \widehat{W}_{rt}^{FLS} \leq \max_{i \in r} x_{it}.$$

Cells with only one distinct OEWS wage cannot be tilted. This may occur because OEWS wages are observed for geographic reporting areas and then assigned to their constituent counties. The effective identifying variation may therefore occur across OEWS reporting areas rather than across individual counties.

7. Separating Sampling Composition from Wage-Concept Differences

The discrepancy

$$\widehat{W}_{rt}^{FLS} - \sum_{i \in r} q_{irt} x_{it}$$

need not consist entirely of random sampling composition. More generally, it contains

$$\begin{aligned} \text{FLS-OEWS discrepancy} = \ & \text{FLS sampling composition} \\ & + \text{FLS-OEWS population differences} \\ & + \text{occupation and hours differences} \\ & + \text{OEWS geographic measurement error} \\ & + \text{reporting, editing, and nonresponse error.} \end{aligned}$$

OEWS and the FLS do not measure identical populations. OEWS is establishment based and reports wages for detailed occupations within OEWS geographic areas. The FLS covers workers directly hired by farms and constructs a field-and-livestock wage using its own worker categories, employment counts, hours, and expansion weights. Systematic differences between these concepts can be absorbed with region effects, year effects, and flexible controls, but idiosyncratic differences may remain.

For that reason, the contemporaneous entropy-calibrated weights cannot automatically be interpreted as pure sampling shocks.

8. Avoiding Mechanical Use of the AEWR Target

If the field-and-livestock FLS wage used to set the AEWR is also used to choose λ_{rt}, an instrument constructed from the resulting weights contains the endogenous AEWR-setting wage by construction. This mechanically strengthens the first stage but creates a circularity problem.

The preferred approach is to infer realized geographic representation from auxiliary FLS moments that are generated by the same survey draw but do not include the field-and-livestock wage used to set the AEWR. Potential auxiliary moments include:

- FLS employment totals by state or region;
- employment by field, livestock, supervisory, and other-worker categories;
- quarterly employment patterns;
- wages for worker categories not used to set the AEWR;
- state-level FLS estimates where available;
- sample sizes, usable response counts, or response rates by state and sampling stratum.

Let $m = 1, \dots, M$ index auxiliary variables, let z_{mit} denote the county analogue of auxiliary variable m, and let Z_{mrt}^{FLS} denote its published FLS regional value. A generalized calibration would choose a low-dimensional vector λ_{rt} satisfying

$$\sum_{i \in r} \omega_{irt}(\lambda_{rt}) z_{mit} = Z_{mrt}^{FLS}, \qquad m = 1, \dots, M,$$

with

$$\omega_{irt}(\lambda_{rt}) = \frac{q_{irt} \exp(\lambda_{rt}' \mathbf{z}_{it})}{\sum_{j \in r} q_{jrt} \exp(\lambda_{rt}' \mathbf{z}_{jt})}.$$

These auxiliary moments reveal which parts of the region received unusually high effective representation. The inferred weights can then be applied to OEWS field-and-livestock wages without using the AEWR-setting wage as a calibration target.

9. Constructing the Leave-Subregion-Out Instrument

After estimating the implied realized weights, counties within each AEWR region are partitioned into dissimilarity subregions. Let $d(i)$ denote county i's subregion, and let target commuting zone k belong to subregion $d(k)$.

The leave-subregion-out wage instrument is

$$Z_{krt} = \frac{\sum_{i \in r:\, d(i) \neq d(k)} \hat{\omega}_{irt} x_{it}}{\sum_{i \in r:\, d(i) \neq d(k)} \hat{\omega}_{irt}}.$$

The exclusion of the target subregion reduces the mechanical contribution of the target commuting zone and economically similar neighboring counties to the donor wage. Under the preferred construction, $\hat{\omega}_{irt}$ is inferred from auxiliary FLS moments that exclude the AEWR-setting field-and-livestock wage.

For an outcome measured in year t, the instrument should follow the timing of the AEWR-setting process. If the AEWR in year t is based on the FLS wage in year $t-1$, the corresponding donor wage and inferred sampling weights should also refer to FLS wage year $t-1$. Specifications using changes must preserve this lag structure explicitly.

10. Empirical Specifications

Several versions should be estimated and reported in increasing order of reliance on the inverse-weight model.

10.1 Expected-weight baseline

Use BEA farm-employment weights without calibration:

$$Z_{krt}^{BEA} = \frac{\sum_{i \in r:\, d(i) \neq d(k)} q_{irt} x_{it}}{\sum_{i \in r:\, d(i) \neq d(k)} q_{irt}}.$$

This measure is predetermined with respect to the FLS sampling realization.

10.2 Design-based prior

Combine BEA employment and agricultural land weights to approximate the list- and area-frame structure. This specification remains independent of the realized FLS wage.

10.3 Auxiliary-moment calibration

Infer the geographic tilt using FLS employment, non-AEWR occupation, response, or sampling moments and then apply the weights to OEWS field-and-livestock wages. This is the preferred approximation to the realized sampling-composition instrument.

10.4 Contemporaneous AEWR calibration

Calibrate directly to the contemporaneous FLS field-and-livestock wage. This version should be treated as an AEWR-implied geographic tilt and as an upper bound on attainable first-stage fit, not as the preferred excluded instrument.

11. Validation and Diagnostics

The construction should report the following diagnostics for every region-year:

1. The BEA-weighted OEWS wage and its difference from the FLS target.
2. Whether the target lies within the support of observed OEWS wages.
3. The estimated tilting parameter or parameters.
4. The effective sample size of the calibrated weights,

$$N_{rt}^{eff} = \frac{1}{\sum_i \hat{\omega}_{irt}^2}.$$

5. The maximum ratio $\hat{\omega}_{irt}/q_{irt}$ and the distribution of weight changes.
6. The number and prior mass of counties with missing OEWS wages.
7. The number of distinct OEWS wage values within the calibration cell.
8. The sensitivity of the weights to dropping individual counties or OEWS reporting areas.
9. The correlation of the inferred tilt with observable agricultural shocks, including crop prices, weather, employment, and H-2A exposure.
10. First-stage strength separately for the BEA baseline, auxiliary-moment calibration, and contemporaneous exact-fit benchmark.

Counties with missing OEWS wages should not simply be discarded and the remaining counties renormalized. Their prior mass must either be retained using a documented imputation, assigned an appropriate OEWS area or state wage, or treated through an explicit partial-calibration procedure.

12. Data Needed for Stronger Identification

The most direct recovery of realized sampling contributions would require FLS microdata or administrative sample information containing farm location, sampling stratum, initial selection weight, response status, nonresponse adjustment, reported worker hours, and final expansion weight. These data are generally not present in published FLS tables.

Short of microdata, useful aggregate information would include, by survey wave and year:

- the number of population farms, sampled farms, and usable responses by state and worker-size stratum;
- list-frame and area-frame sample counts;
- stratum-specific sampling fractions and response adjustments;
- state-by-category expanded employment or hours;
- replicate weights or stratum-level variance contributions.

Such information would add independent restrictions on the realized weight vector and reduce reliance on the minimum-divergence assumption. If it is not publicly released, it may be obtainable through a request to NASS for nondisclosive state-by-stratum aggregates.

13. Interpretation

The empirical design is best understood as an attempt to use independent county wage data to reconstruct how random FLS sampling and response patterns changed the geographic composition of the wage estimator. The BEA weights describe expected representation. Auxiliary FLS moments reveal deviations from that representation. OEWS wages translate the inferred deviations into a predicted AEWR sampling shock.

The strongest claim supported by aggregate data is not that the actual county sampling weights have been observed. Rather, the method estimates a low-dimensional, minimum-divergence approximation to the realized geographic tilt. Its credibility depends on three conditions:

1. the BEA or design-based prior adequately describes expected geographic contribution;
2. the auxiliary calibration moments respond to realized FLS representation in a way that can be matched to county data; and
3. the calibration excludes the contemporaneous field-and-livestock wage used to set the AEWR.

Under those conditions, the leave-subregion-out weighted OEWS wage provides a plausible proxy for the random component of the AEWR generated by the realized FLS sample. Direct calibration to the AEWR remains useful as a descriptive decomposition and first-stage benchmark, but it should not be the primary excluded instrument.

References and Methodology Sources

- USDA National Agricultural Statistics Service, Agricultural (Farm) Labor Survey.
- USDA National Agricultural Statistics Service, Farm Labor Methodology and Quality Measures.
- USDA National Agricultural Statistics Service, Farm Labor Methodology and Quality Measures archive.